

Experiment 08: Develop an application that uses GUI components, Fonts and colors

PART A

A.1 Aim: To develop an application that uses GUI components, Fonts and colors.

A.2 Objectives: To introduce students with various tools like Android Studio, NS2, Wire-shark, Cisco packet tracer, WAP supported browser etc.

A.3 Outcome:

After successful completion of this experiment students will be able to Develop an application that uses GUI components, Fonts and colours

A.4 Theory:

SOFTWARE:

- Android Studio
- The Android SDK (Starter Package)
- Gradle
- Java Development Kit (JDK) 5

DESCRIPTION:

- 1 Open android studio and select new android project .
- 2 Give project name and select next 3
Choose the android version.
- 4 Enter the package name. package name must be two word separated by comma and click finish
- 5 Go to package explorer in the left hand side and select our project.
- 6 Go to res folder and select layout. Double click the main.xml file 7 Now you can see the Graphics layout window.

Source code:
CodeforActivity_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"android:orient
ati on="vertical"

android:layout_width="match_parent"android:layout_height="mat ch_parent">

<TextViewandroid:id="@+id/textVi
ew"android:layout_width="match_
parent"

android:layout_height="wrap_content"
a
ndroid:layout_margin="30dp"android:
g ravity="center"android:text="Hello
World!"android:textSize="25sp"androi
d:textStyle="bold" />
<Button

android:id="@+id/button1"android:lay
out_width="match_parent"android:layo
ut_height="wrap_content"android:layo
ut_margin="20dp"android:gravity="cent
er"android:text="Change font size"android:textSize="25sp"
/>

<Button

android:id="@+id/button2"android:lay
out_width="match_parent"android:lay o
```

```

        ut_height="wrap_content"android:layo
        u

        t_margin="20dp"android:gravity="cent
        er"android:text="Change color"android:textSize="25sp"
    />
< /LinearLayout >

```

CodeforMainActivity.java:

```

ackagecom.example.exno1;

importandroid.graphics.Col
or; import

android.support.v7.app.AppCompatActivity
; im port android.os.Bundle; import
android.view.View;import
android.widget.Button;impo rta
ndroid.widget.TextView;

publicclassMainActivityextendsAppCompatActivity
{
    intch=1;floa
    t
    font=30;@
    O verride

    protectedvoidonCreate(Bundle savedInstanceState)

    { super.onCreate(savedInstanceState);setContentView(R.layout.a ctivity_main); final
    TextView t= (TextView)

        findViewById(R.id.textView);Button b1= (Button)

        findViewById(R.id.button1);b1.setOnClickListener

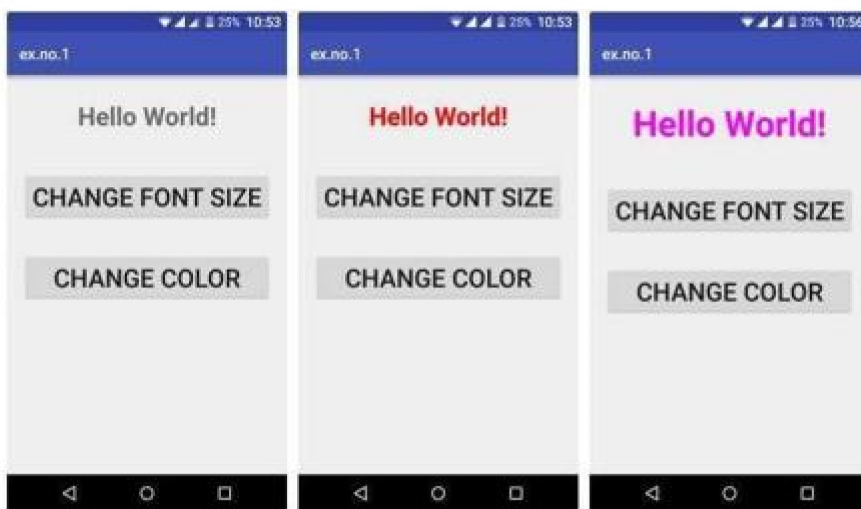
        ( ne wView.OnClickListener){

```

```
@Override public void  
onClick(View  
  
v)  
{t.setTextSize(font); font  
= font +  
  
5;if (font  
== 50)font  
=30;  
}  
});  
Button b2= (Button)  
findViewById(R.id.button2);b2.setOnClickListener(  
er(new View.OnClickListener(){  
  
@Override  
public void onClick(View  
v)  
  
{switch(ch){ case1:  
  
t.setTextColor(Color.RED); break  
  
case2:  
t.setTextColor(Color.GREEN); break;  
  
case3:  
t.setTextColor(Color.BLUE)  
; break;  
  
case4:  
t.setTextColor(Color.CYAN); break;  
  
case5:  
t.setTextColor(Color.YELLOW); break;
```

```
case6: t.setTextColor(Color.MAGENTA);
    break;
}
}
});
}
}
ch++;
if ( ch == 7 ): ch
    == 1;
```

Output:



PART B

(PART B: TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case there is no Black board access available)

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Class : TE Comps C	Batch: C-3
Date of Experiment:24.03.25	Date of Submission:31.03.25
Grade:	

B.1 Software Code written by student/steps:

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <!-- Permission to access the internet -->
    <uses-permission android:name="android.permission.INTERNET"/>

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="GUI App"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.GUIApp">

        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

```
        </activity>

    </application>

</manifest>
```

MainActivity.java

```
package com.example.application; // Ensure this matches your project package

import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    int ch = 1;
    float font = 30f;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        TextView t = findViewById(R.id.textView);
        Button b1 = findViewById(R.id.button1);
        Button b2 = findViewById(R.id.button2);

        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                t.setTextSize(font);
                font += 5;
                if (font > 50) font = 30;
            }
        });

        b2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                switch (ch) {
                    case 1: t.setTextColor(Color.RED); break;
                    case 2: t.setTextColor(Color.GREEN); break;
                    case 3: t.setTextColor(Color.BLUE); break;
                    case 4: t.setTextColor(Color.CYAN); break;
                    case 5: t.setTextColor(Color.YELLOW); break;
                    case 6: t.setTextColor(Color.MAGENTA); break;
                }
                ch++;
            }
        });
    }
}
```

```

        if (ch > 6) ch = 1;
    }
    });
}
}

```

Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

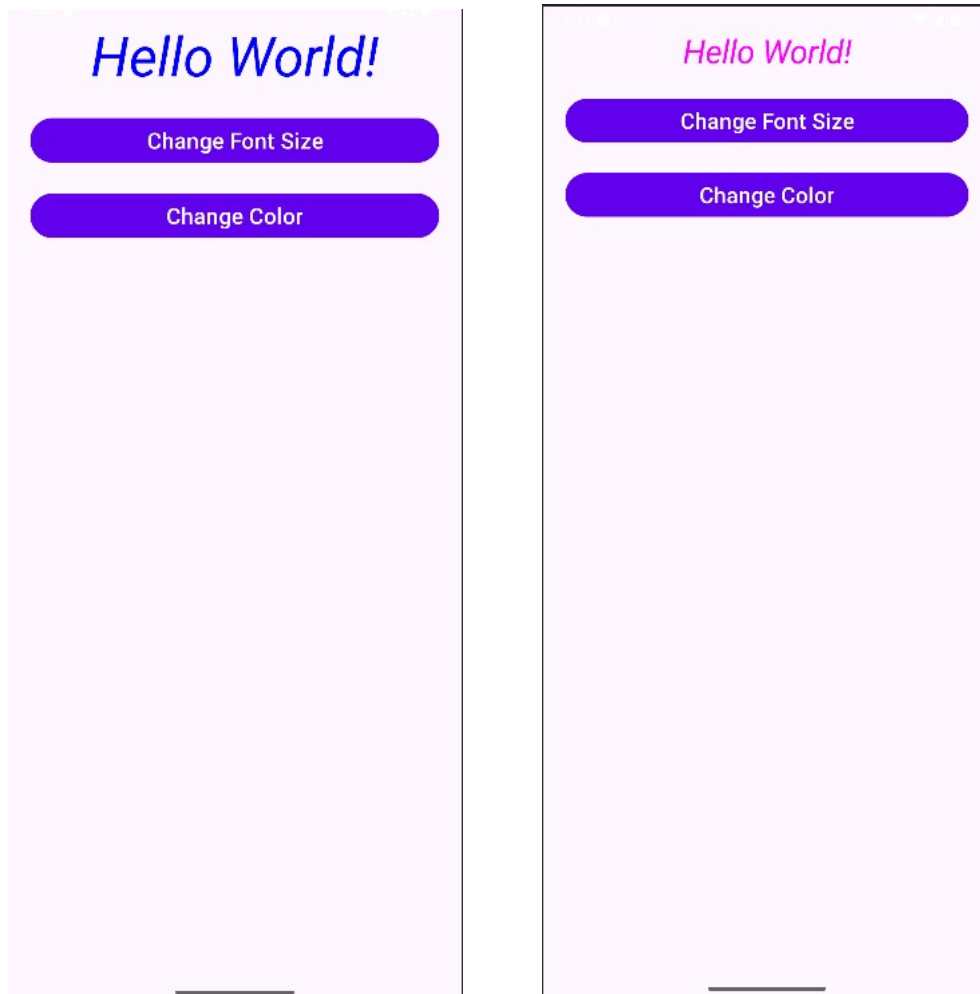
    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:text="Hello World!"
        android:textSize="25sp"
        android:textStyle="italic" />

    <Button
        android:id="@+id/button1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Change Font Size"
        android:textSize="20sp" />

    <Button
        android:id="@+id/button2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Change Color"
        android:textSize="20sp" />
</LinearLayout>

```

B.2 Input and Output:



B.3 Observations and learning:

- The GUI components responded as expected to user interactions.
- Fonts and colors were applied without any rendering issues.
- The app maintained a clean and intuitive layout across different screen sizes.
- Performance remained smooth, indicating optimized design practices.

B.4 Conclusion:

The application was successfully developed using GUI components such as buttons, text views, and input fields. Various fonts and colors were applied to enhance the visual appearance of the app. The user interface was designed to be interactive and visually appealing, demonstrating effective use of Android's UI customization capabilities.

B.5 Question of Curiosity

1) Explain different steps required to build up this project?

Steps to Build an Android Application Using GUI Components, Fonts, and Colors

1. Setup the Environment:

- Install Android Studio and configure necessary SDK tools.
- Create a new Android project using an empty activity template.

2. Design the Layout:

- Use XML to create the UI using components like `TextView`, `Button`, `EditText`, etc.
- Arrange components using `LinearLayout`, `RelativeLayout`, or `ConstraintLayout`.

3. Apply Fonts and Colors:

- Add custom fonts by placing `.ttf` files in the `res/font` folder.
- Define colors in `res/values/colors.xml`.
- Apply the colors and fonts using `android:textColor`, `android:fontFamily`, and `android:background`.

4. Implement Functionality:

- Use Java or Kotlin in the `MainActivity` to handle button clicks and input events.
- Link UI components using `findViewById()`.
- Update text views and display results based on user input.

5. Testing and Debugging:

- Run the application on an emulator or a physical device.
- Debug and fix any issues using Logcat.

6. Finalization:

- Adjust UI design for different screen sizes using responsive layouts.
- Perform testing for UI consistency and functionality.

- Build and generate the APK for deployment.